

Computational Design and Fabrication of 3D Printed Zippers Connecting 3D Textile Structures

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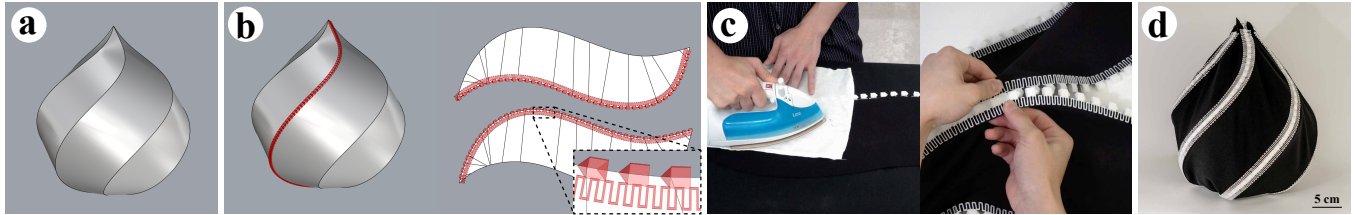


Figure 1: Our fabrication pipeline. **a:** The input 3D model divided into developable patches. **b:** The output model of the curved zipper attached to the target model and its unrolled patterns. **c:** Thermally bonding 3D printed zippers on textiles and manually connecting them. **d:** The fabricated object. Note that we inserted a mechanical support for the photo to reduce the gravity force.

Abstract

Zippers have long been used for connecting and disconnecting two textiles repeatedly and easily. However, many industrial zippers cannot retain their shape after assembly, limiting the potential use of connecting soft textiles to form 3D shapes with large and various curvatures. Thus, we present a method to design and fabricate interlocking 3D printable zippers that can encode post-assembly shapes. The user first gives a target 3D shape divided into developable patches (i.e., curved surfaces unrollable like paper). Then, our design software built on Rhino/Grasshopper computes an interlocking zipper on the boundary curve of the patches. The user 3D prints the zipper in a flat state and welds it to the edge of the textiles by thermal bonding, which can zip into a target 3D shape. In this demo, we report our method and exhibit design examples of 3D printed zippers.

CCS Concepts

• **Applied computing** → **Computer-aided design**; • **Hardware** → **Printers**; *Emerging tools and methodologies*.

Keywords

3D printing, Computational Design and Fabrication, Zippers

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1 Introduction

From air hose valves to USB cables, connectors are ubiquitous in our lives. Assembling small components into a larger structure with connectors has three merits rather than fabricating a final shape at once: (1) Objects can be *detachable* for user interaction (e.g., a zip-up hoodie is easier to wear than a pullover); (2) 3D structures can be *deployable* from easy-to-fabricate flat structures like paper craft [3]; (3) Connected objects are *deformable* and can be larger in volume after fabrication (e.g., 3D printed objects can be larger than 3D printers by deforming hinge structures after fabrication [2, 5]).

Among many types of mechanical connectors, we focus on zippers. Zippers are promising because they satisfy *detachability*, *deployability*, and *deformability* by user's easy manipulation. Previously, Schüller et al. [4] proposed a method to decompose a 3D surface into developable patches assembled with industrial zippers. However, such zippers are not mechanically stable in the zipped-up state; they catastrophically fail to retain their target shape even for a simple cylinder (Figure 3).

We present a method to design and fabricate interlocking 3D printable zippers that can encode their post-assembly shape. Figure 1 shows the overview of our fabrication pipeline. Given a 3D surface divided into developable patches, the system implemented on Grasshopper generates a zipper following the boundary curve of the patches. The user can weld printed zippers on the flat textiles by thermal bonding and zip them up to form the final shape. The zippers in the interlocked state are mechanically stable in target shapes as long as the weight is within their structural limit (Figure 3). In this demo, we demonstrate the design principle and examples.

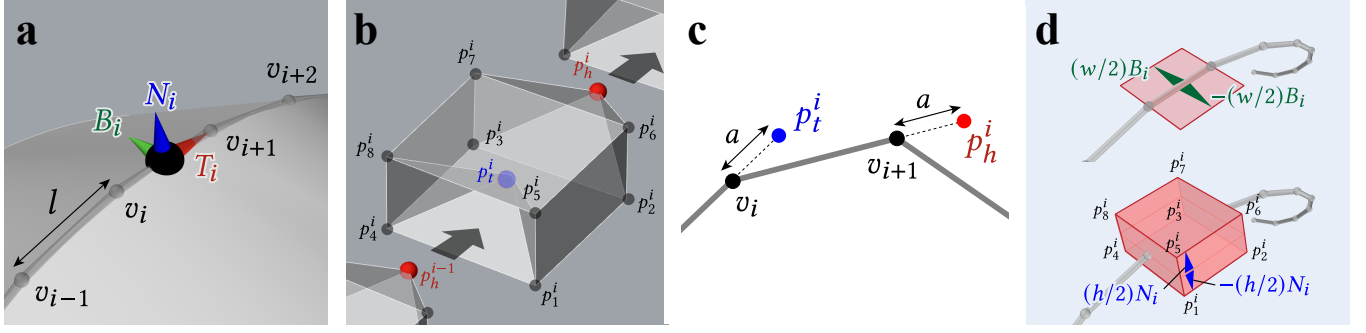


Figure 2: The geometric pipeline to design our zipper elements.

2 Method

2.1 Design of Zipper Elements

Given developable patches forming a 3D surface, our design software first extracts all the boundary curves between the patches (Figure 1b left). After this procedure, the system refines the curve into a polyline that consists of n nodes $v_1, \dots, v_n$ and $n - 1$ segments $v_1v_2, \dots, v_{n-1}v_n$ where every segment has a constant length l (Figure 2a). For each segment v_iv_{i+1} , we denote the unit tangent vector as T_i , the unit normal vector of the surface on the segment midpoint as N_i , and their cross product as B_i (Figure 2a).

For each segment v_iv_{i+1} , we wish to assign a zipper element shown in Figure 2b parametrized by its head p_h^i , tail p_t^i , and other points $p_1^i, \dots, p_8^i$. When assembled, the zipper elements are interlocked to each other by inserting the head p_h^i of a segment v_iv_{i+1} into the tail p_t^{i+1} of the next segment $v_{i+1}v_{i+2}$. To achieve this, we set

$$p_h^i = v_{i+1} + aT_i, \quad (1)$$

$$p_t^i = v_i + aT_{i-1}, \quad (2)$$

where a is the size of the head/tail (Figure 2c). Also, to obtain $p_1^i, \dots, p_8^i$, we run the following procedure:

1. Sweep the whole polyline towards B_i and $-B_i$ for a width of $w/2$ each. This gives a strip of quadrilateral faces (Figure 2d top).

2. Sweep the whole strip towards N_i and $-N_i$ for a width of $h/2$ each. This gives a sequence of hexahedral blocks (Figure 2d bottom), and we assign the eight points on each block to $p_1^i, \dots, p_8^i$.

In our experiment, we set $l = 10$ mm, $a = 5$ mm, $w = 10$ mm, and $h = 5$ mm.

Given the geometry of these zipper elements, the system unrolls all of them onto the 2D plane and connects them with a meandered structure (see Figure 1b right).

We note that the user can prepare the input developable surfaces, for example, by a Rhino command DevLoft.

2.2 Fabrication

After designing the zipper elements, the user 3D prints them using a standard FDM 3D printer with a PLA filament. Our system generates a meandered flexible structure beside the zipper elements (Figure 1b right), and the user can weld this part to textile by thermal bonding (Figure 1c).

One important property to consider when we choose the appropriate textile is its stretchability. If the textile is not stretchable at all, the misalignment error during the welding process accumulates as we zip up, making it difficult to close the zipper completely. Another important property required is heat durability, as otherwise, the textile will be damaged after thermal bonding. In our experiment, we used a stretchable polyester-polyurethane fabric.

3 Results, Discussion, and Conclusion

As shown in Figures 1 and 3, our method succeeded in reproducing the target shape thanks to our interlocking structure. In the demo, we will exhibit multiple design examples made by our zippers.

Meanwhile, we still have several problems to solve. First, the assembly process requires manual labor to interlock all the pieces (Figure 1c right). To achieve faster assembly, we plan to introduce the computational design of sliders, which is a common practice for industrial zippers. Also, the thermal bonding process consumes time and limits the material selection. We will consider 3D printing zippers directly onto the textile or 3D printing the textile [1, 6]. Finally, the mechanical test of the zippers against the gravity force will be important to explore the possible form factor.

We hope that our 3D printable zipper will work as a new toolbox for fabricating detachable, deployable, and deformable objects.

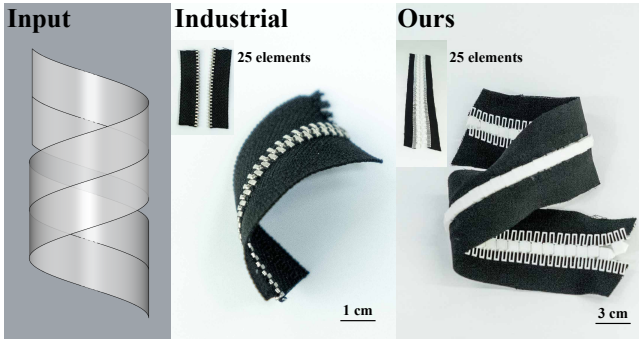


Figure 3: Unlike industrial zippers (center), our interlocking zipper is mechanically stable in the final shape (right).

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